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Measurement of electric flux emission: a new diagnostic for the dense plasma focus

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Abstract

This paper presents a novel diagnostic technique for investigating the symmetry of the plasma formation process in the Dense Plasma Focus (DPF) device. The DPF, a scalable laboratory plasma device developed in the 1960s, has been extensively studied with various gas fills and considered for a broad range of applications—including commercial fusion energy, short-lived radioisotope production for medical imaging and brachytherapy, nano-scale industrial coatings, and deep-space propulsion. However, achieving reliable and reproducible operation of the DPF remains a significant challenge. Plasma initiation occurs via a surface discharge across an insulating sleeve between the two coaxial electrodes of the DPF device. Typically, a series of ‘training’ shots is required before symmetric plasma formation is achieved, yet the mechanisms underlying this process remain poorly understood. Numerical simulations have revealed that plasma formation involves complex atomic and radiative phenomena spanning wide spatial and temporal scales, rendering predictive modelling nearly impossible. The diagnostic technique introduced here provides new empirical insights—unavailable through existing methods—into the physics and phenomenology of plasma formation. It utilizes three symmetrically arranged, absolutely calibrated, and identical D-dot probes to measure the electric flux emitted by the plasma, potentially offering a pathway toward more reproducible DPF operation through a better understanding of the plasma formation process.

1. Introduction

The Dense Plasma Focus (DPF) [1–3] is a relatively simple and scalable laboratory plasma device extensively studied since its discovery in 1962 in the USSR [4] and in 1964 in the USA [5]. It consists of two coaxially mounted cylindrical electrodes separated by an insulator sleeve, placed inside a vacuum chamber. The vacuum chamber is filled with a gas at few millibars pressure. A fast, high voltage, high current capacitor bank is switched across the electrodes. This leads to a surface plasma discharge over the insulator that carries current between the electrodes. The magnetic force between the current in the plasma and the magnetic field created by it, acts perpendicular to the plasma. Under the action of this force, the plasma changes its shape and accelerates along the central electrode. At the end of this electrode, it turns and collides with itself, transforming its fluid kinetic and magnetic energy into particle kinetic energy of a non-thermal, turbulent, magnetized plasma.

This rapid convergent flow towards the axis is the main distinguishing feature of the Dense Plasma Focus. Its stagnation at the axis creates a hot and dense magnetized plasma object—usually called ‘the pinch’. The more symmetric the convergent flow, the hotter and denser—and consequently more useful—the pinch is. It disintegrates by outflow and instabilities in a very complex manner [3]. Destruction of the magnetic flux embedded within the pinch produces energetic electrons and ions which produce intense radiation and cause nuclear reactions. It also results in a characteristic sharp dip in the current derivative signal that is often named

as ‘the singularity’ in research literature. It provides a unique time reference and serves as a confirmation of adequately symmetric convergent flow.

The plasma focus has been operated with a wide variety of gases resulting in a range of novel properties. When operated with deuterium or deuterium-tritium, it produces an intense burst of fusion neutrons (2.5 MeV for deuterium, 14 MeV for deuterium tritium) with a duration of a few tens to hundreds of nanoseconds [1–3]. This has been utilized to build a dedicated large DPF facility for using high fidelity pulsed neutron radiography in nuclear weapons research based on subcritical implosions [6]. The fusion reaction yield scales as the fourth power of current [1–3]. Efforts at utilizing this scaling for commercial production of fusion energy are under way [7–9]. Space propulsion concepts for deep space and interplanetary travel based on the DPF are reported to have unique advantages over other alternatives [10, 11]. Significant quantities of short-lived radioisotopes of interest in medical imaging and brachytherapy have been produced with special gas mixtures [12, 13]. Industrial grade nano-scale coatings suitable for production of next generation information storage media have been demonstrated [14] as part of a wider research program on applications of the DPF for nanotechnology [15, 16].

The DPF is, therefore, certainly a very valuable device. Unfortunately, in terms of reliable and reproducible operation, it leaves much to be desired. Use of ultra-high vacuum technology appears [17] to mitigate, but not eliminate, this problem. Reproducibility is related with the plasma formation process. The insulator needs to undergo a conditioning process involving a few training shots before the desirable operating regime with good plasma symmetry manifests itself. The nature of this process is poorly understood and might hold the key to a reliable and reproducible operation.

Our previous work [18–20] on numerical simulations of the plasma formation process using PIC-MCC (Particle In Cell - Monte Carlo Collisions) method has revealed that it involves atomic and radiative phenomena having a wide range of spatial and temporal scales [21–23] which makes predictive modelling very difficult, if not impossible. We have therefore embarked on an experimental program to generate a systematic understanding of the role of the training shots in the formation process [24].

It is empirically known [1, 2] that the device works best when a thin, azimuthally uniform plasma sheath is formed. The first requirement of this program, therefore, is generation of data on the symmetry of the plasma formation and propagation process. This is proposed to be achieved by measurement of electric flux emission from the plasma at different azimuthal positions.

Most laboratory plasmas are quasi-neutral in the sense that the absolute difference in the number densities n_i and n_e of ions and electrons is much less than their average value n . However, as a rule, they are not charge-neutral. The charge density ρ_e on a finite plasma is related to the electric field $\vec{E}$ and permittivity of vacuum ϵ_0 by Gauss’s Law

$$\rho_e = \epsilon_0 \vec{\nabla} \cdot \vec{E} \quad (1)$$

along with the Generalized Ohm’s Law, which determines the electric field in terms of fluid velocity $\vec{v}$, magnetic field $\vec{B}$, resistivity η , current density $\vec{J}$, electronic charge e and electron pressure gradient $\vec{\nabla} p_e$:

$$\vec{E} = -\vec{v} \times \vec{B} + \eta \vec{J} + (en)^{-1}(\vec{J} \times \vec{B} - \vec{\nabla} p_e) \quad (2)$$

Substitution of expression (2) into (1) results in the term $-\epsilon_0 \vec{\nabla} \cdot (\vec{\nabla} p_e / en)$ which is always nonzero for a finite plasma. A finite plasma must therefore emit electric flux. This simply means that a floating conductor in the vicinity of the plasma must experience an induced charge proportional to the electric flux incident on it. If this charge were to be transported via a resistor to the laboratory ground, the resulting current would generate a voltage signal across the resistance that can serve as a diagnostic for the electric flux emitted by the plasma and intercepted by the conductor. Placing a number of such conductors around the plasma, one could obtain some information about whether the plasma emits equal amounts of electric flux in different azimuthal directions. These data can be compiled in correlation with various operating parameters of the device from which inferences could be drawn and theoretical models could be constructed and tested against further investigations.

This paper describes measurement of electric flux emission as a new diagnostic for studying the DPF. It is an adaptation of the d-dot probe technique [25] used in pulsed power diagnostics. Section 2 describes the experimental hardware and theoretical analysis. Section 3 describes calibration experiments. Section 4 describes preliminary results with the DPF. Section 5 concludes the paper with a summary.

2. Experimental hardware and theory of measurement

Figure 1 shows a schematic of the hardware configuration. A hexagonal 3D printed component acts as a support for up to 6 SMA type right angle PCB adaptors (SMA PCB Through Hole Right Angle Jack 50 Ohm).

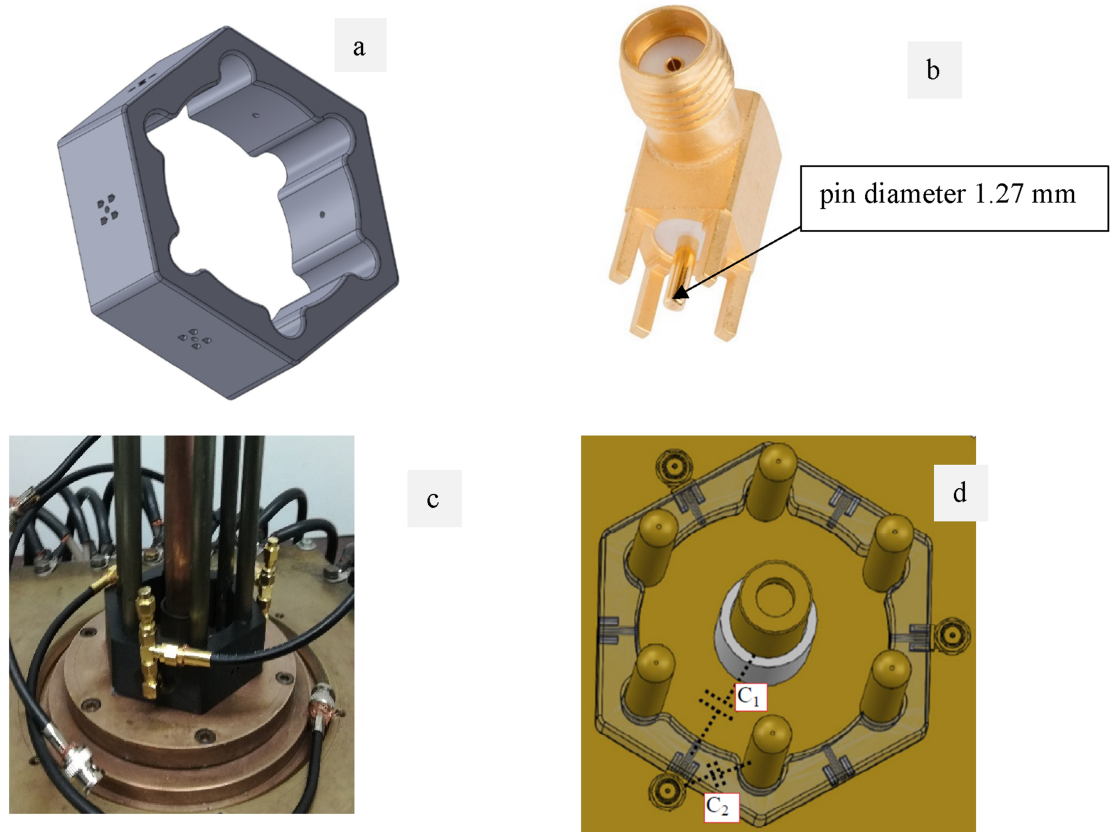


Figure 1. A hexagonal 3D printed component (figure 1(a)) acts as a support for up to 6 SMA type right angle PCB adaptors (shown in figure 1(b)) which can be placed symmetrically around the plasma focus between the gaps in the squirrel cage cathode (shown in figure 1(c)). An electric charge is induced on the central pin of the adaptor because of voltage on the anode, represented as a stray capacitance C_1 (shown in figure 1(d)). There is also a stray capacitance C_2 between the connector and the cathode rods. Figure 1(b) is reproduced from the website (<https://www.amphenolrf.com/901-143.html>).

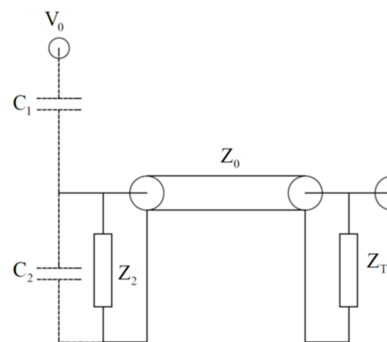


Figure 2. Circuit diagram. Capacitance C_1 is the circuit representation of the charge induced on the central pin of the SMA adaptor by a conductor raised to potential V_0 . Capacitance C_2 is the circuit representation of likely electrostatic coupling between the SMA adaptor and the cathode rods. Z_0 is the cable impedance. Z_2 is the termination of the cable at its input end and Z_T that at its oscilloscope end.

The central pin of the SMA adaptor, connected to the oscilloscope input via a coaxial cable, acts as a floating conductor. The coaxial cable is terminated in 50 Ohms at both ends. Electric flux incident on the pin induces a charge that flows to the cable ground through the 50 Ohm termination at the input end of the cable, that ultimately results in a voltage across the input amplifier of the oscilloscope. A theoretical relationship between the two can serve as a basis of measurement of the electric flux.

The hardware configuration of figure 1 can be represented by the circuit diagram shown in figure 2.

The coaxial cable has a characteristic impedance Z_0 , is terminated at the oscilloscope end by impedance Z_T and has the impedance Z_2 across its input end. When the terminating impedance equals the characteristic

impedance, the input impedance of the coaxial cable is known [26] to equal the characteristic impedance: $Z_{input} = Z_0$.

The Thevenin equivalent voltage source in figure 2 is

$$V_{Th} = V_0 \frac{Z_{||}}{Z_{||} + 1/j\omega C_1}; \quad Z_{||} = \frac{Z_2}{j\omega Z_2 C_2 + 1} \quad (3)$$

in series with a Thevenin equivalent impedance

$$Z_{Th} = \frac{Z_2}{j\omega Z_2 (C_1 + C_2) + 1} \quad (4)$$

The voltage developed across the input impedance of the coaxial cable is therefore

$$\begin{aligned} V_{input} &= V_{Th} \frac{Z_{input}}{Z_{input} + Z_{Th}} \\ &= V_0 j\omega C_1 Z_2 \frac{Z_{input}}{Z_{input} (j\omega Z_2 (C_1 + C_2) + 1) + Z_2} \end{aligned} \quad (5)$$

In our diagnostic, $Z_{input} = Z_2 = Z_0 = 50\Omega$ and for a frequency range where $\omega \ll 1/Z_0(C_1 + C_2)$ (5) simplifies to

$$V_{input} = \frac{1}{2} V_0 j\omega C_1 Z_0 \quad (6)$$

At the output of the cable, the input signal is divided between the cable impedance and the termination impedance so that the voltage at the oscilloscope is

$$V_{osc} = \frac{1}{4} j\omega C_1 V_0 Z_0 \quad (7)$$

The quantity $j\omega C_1 V_0$ is the displacement current and $C_1 V_0$ is the charge induced on the pin of the SMA adaptor. The pin has a standard diameter $d = 1.27$ mm. The value of the displacement D at the location of the pin of the SMA connector is therefore obtained by the relation

$$D = \frac{C_1 V_0}{\pi (d/2)^2} = \frac{16}{Z_0 \pi d^2} \int dt V_{osc} \quad (8)$$

The electric displacement is a local property of the electromagnetic field emitted by the plasma that is highly sensitive to its geometry—the charge on the portion of the plasma nearest to the pin produces the most displacement. If symmetrically placed d-dot probes do not give similar signals, the emission of electric flux by the plasma is not symmetric and the plasma is not capable of transporting the current delivered by the capacitor bank to its pinch phase in a stable manner.

3. Calibration experiments

The d-dot probe is self-calibrated by its very nature. However, its use for determination of the symmetry of the plasma needs an experimental confirmation of the electrically symmetric placement of the d-dot probes. This confirmation is obtained in the following manner. A test jig is improvised as shown in figure 3.

It supports the central conductor of a coaxial cable at the geometric centre of the hexagon. The other end of the cable is soldered to the output of a high voltage trigger pulse generator [27]. A calibrated oscilloscope probe is connected across a voltage divider built into the output load resistance of the trigger generator, whose resistances have measured values 1.306 kOhm and 13.2 Ohm. The electric flux, obviously emitted symmetrically by the central conductor, induces signals in the three d-dot probes. Figure 4 shows the results.

The maximum values of the integrated d-dot signals are within 3% of the mean value, which is:

$$\int dt V_{osc} = 4 \times 10^{-9} \text{V} \cdot \text{s} \quad (9)$$

The maximum value of the applied voltage is 5.34 kV. The measured value of capacitance C_1 in figure 3 is therefore

$$C_1 = \frac{4}{V_0 Z_0} \int dt V_{osc} \approx 0.006 \text{pF} \quad (10)$$

This value however does not apply to the configuration of figure 1(c) and is of passing interest merely to confirm that the capacitance has the correct ballpark value. The major takeaway from this experiment is the confirmation that the diagnostic has baseline symmetry by itself. Any deviation from this symmetry is therefore solely a property of the electromagnetic environment it is placed in.

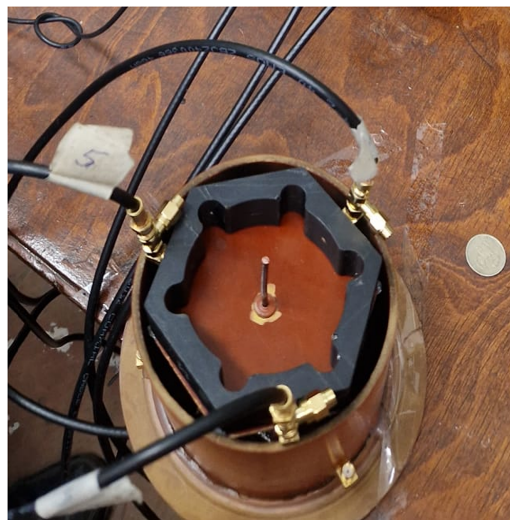


Figure 3. A test jig supports the central conductor of a coaxial cable at the geometric centre of the hexagon. Its other end is soldered to the output of a high voltage trigger pulse generator.

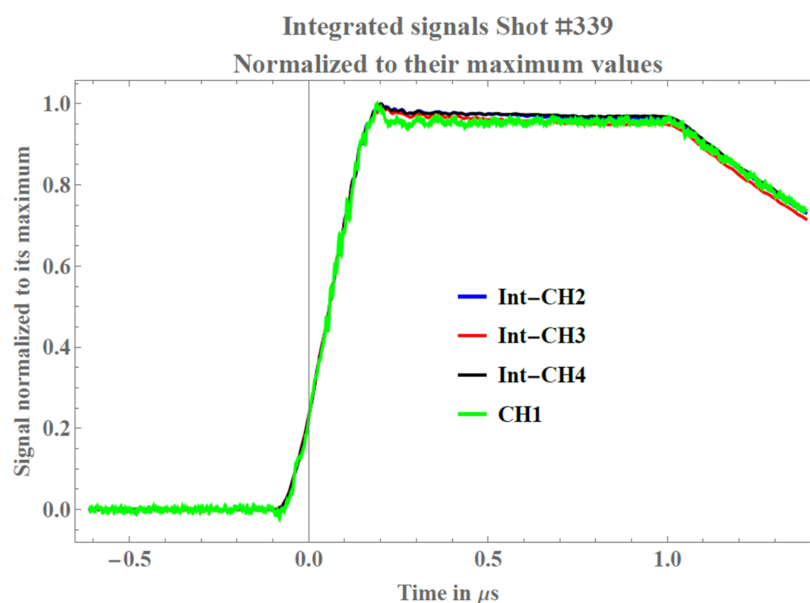


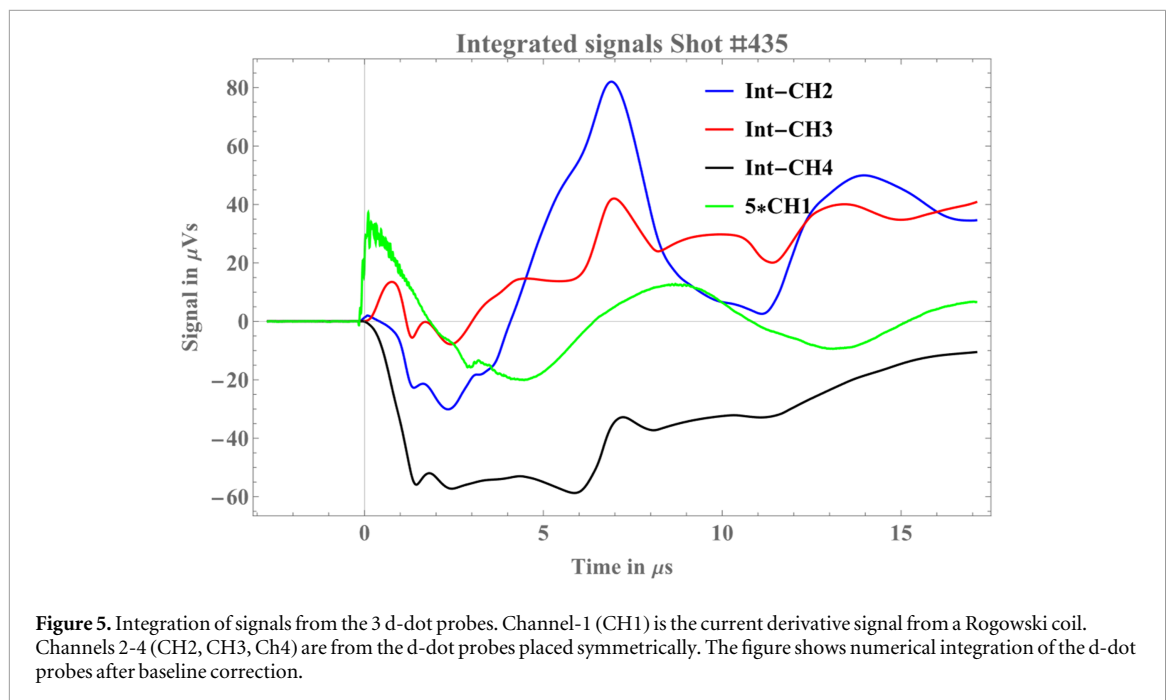
Figure 4. Numerical integration of the three d-dot probe signals CH2, CH3, CH4 compared with the input waveform CH1 measured with a resistive voltage divider. Baseline correction is applied prior to numerical integration. The integrated d-dot signals normalized to their maximum values overlap exactly. The maximum values themselves are within a $\pm 3\%$ range of their mean.

4. Tests with a plasma focus

This diagnostic has been deployed on ‘...the University of Sofia plasma focus, a 3 kJ Mather type device (20 μ F, 40 kV). Its anode is a hollow copper tube 2 cm in diameter and 14.5 cm in length. The cathode consists of six copper rods (0.8 cm diameter, 16 cm length) mounted on a massive cathode base plate on a circle with a 3.5 cm radius. The stainless-steel vacuum chamber has 15.5 cm inner diameter and 35 cm height. This device can be operated with air, argon or deuterium.’ [28]

It was noticed that the integration of signals from the three d-dot probes were highly dissimilar as seen in figure 5.

Doubts about the inherent symmetry of the new diagnostic were resolved as shown in section 3. Therefore, the only possibility that remained was that the source of asymmetry was within the plasma focus, which had been in normal operation for a considerable time. Disassembly of the plasma focus led to the discovery that the hollow copper tube had a deformation which was hidden from the view behind and a little further down from the bottom edge of the insulator (figure 6). This did not affect the pinch formation since the plasma formed on



the insulator was adequately symmetric. The d-dot probes were, however, sensitive to the electric flux incident on them. Since the electric flux emission was dependent on the geometry of the anode, the invisible deformation of the hollow anode revealed itself in the marked asymmetry of the electric flux emission signals.

This deformation is obviously related to the fact that the anode was hollow. It was made from a copper tube. Copper tubes are typically annealed after drawing and are therefore ductile. Any asymmetrical magnetic impulse would thus cause a slight deformation. This would increase the asymmetry in the subsequent shots and amplify the distortion—a slow, solid-state version of a kink instability.

After replacing the hollow copper electrode with a solid brass electrode, the following results (figure 7) were observed:

Numerical integrations of the signals in figure 7 are shown in figure 8.

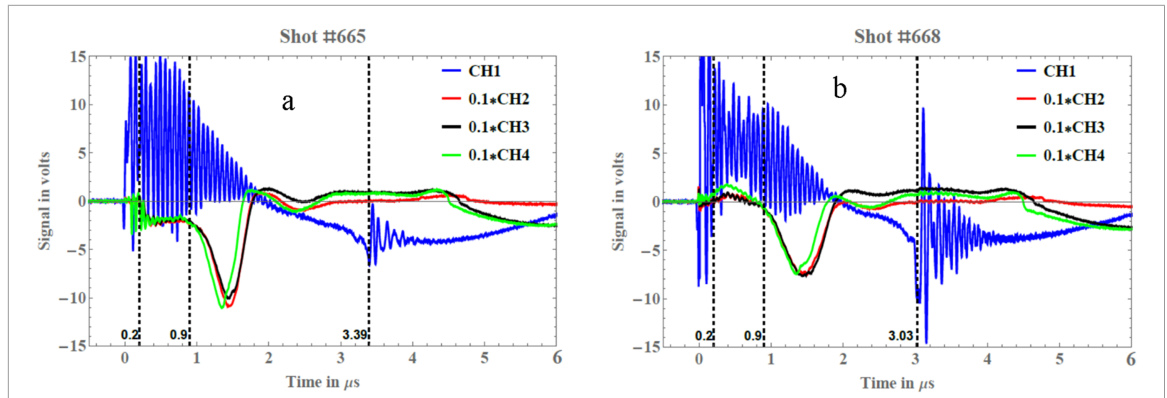


Figure 7. Two examples of d-dot probe signals with a normal plasma focus operation. (a) Shot #665 with 0.95 Torr of Ar and 19.0 kV charging voltage. (b) Shot #668 with 0.83 Torr of Ar and 19.1 kV. All the signals are obtained at a sampling of 1 ns per point and smoothed with 10-point moving average algorithm.

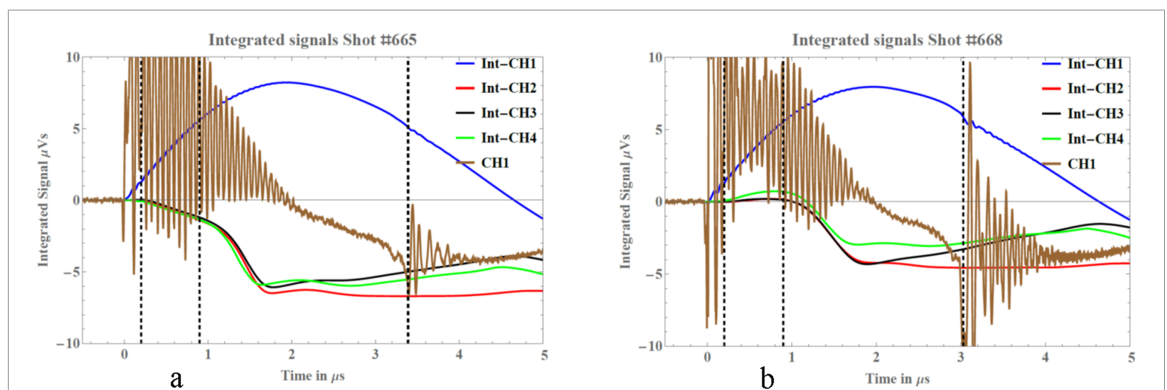


Figure 8. Numerical integrations of signals in figure 7.

The two shots differ in the proximity of the current derivative singularity from the current maximum (zero of the current derivative). Shot #668 is at a lower pressure, hence the plasma velocity is higher and the singularity occurs earlier. It is well known from plasma focus model simulations [29] that the current maximum signifies the end of the rundown phase. Two d-dot signals, CH2 and CH3, follow each other quite well during the rundown phase while the third d-dot signal, CH4, has a marked deviation. This is an indication of adequate, but imperfect, symmetry of the plasma during the rundown phase. The interval between the current maximum and the singularity is the radial phase [29].

The rundown phase corresponds to supersonic motion of the plasma. The major contribution to the electric field must then come from the motional electric field $-\vec{v} \times \vec{B}$. Figure 8(b) shows that the electric flux emission signals from CH2 and CH3 are close to zero up to the second vertical dashed line. This must then signify the plasma formation and build-up phase. Comparison of figures 7 and 8 shows that the plasma is not quite stationary in the time interval indicated by the first two vertical dashed line, even though it is not moving supersonically. This is expected since increase in its density and pressure must lead to sub-sonic expansion. However, the behaviour of the electric flux emission signals is quite different for the two shots, indicating presence of more complex processes in the formation phase.

Figure 9 shows an expanded view of shot #665 with the baseline of the current derivative signal shifted vertically. Note that the d-dot signal from CH4 starts earlier than those from CH2 and CH3, showing that the surface discharge begins earlier in its vicinity, which probably corresponds to a physical asymmetry in the plasma focus near the insulator. Also noteworthy is the fact that all the three d-dot probes exhibit oscillations which are frequency and phase correlated with the oscillations in the current derivative signals. This fact is very significant. The current derivative signal is measured outside the plasma focus chamber, ostensibly derived from the rate of change of the azimuthal magnetic field produced by the current. The d-dot signals are, on the other hand, immune to magnetic field. This correlation indicates that the Rogowski coil is inadequately shielded from the electric flux emitted by the anode outside the plasma focus and incident on the coil.

It is also important to note that the CH2 and CH3 signals are identical in the formation phase (figure 9) and much of the rundown phase (figures 7 and 8) but deviate in the radial phase.

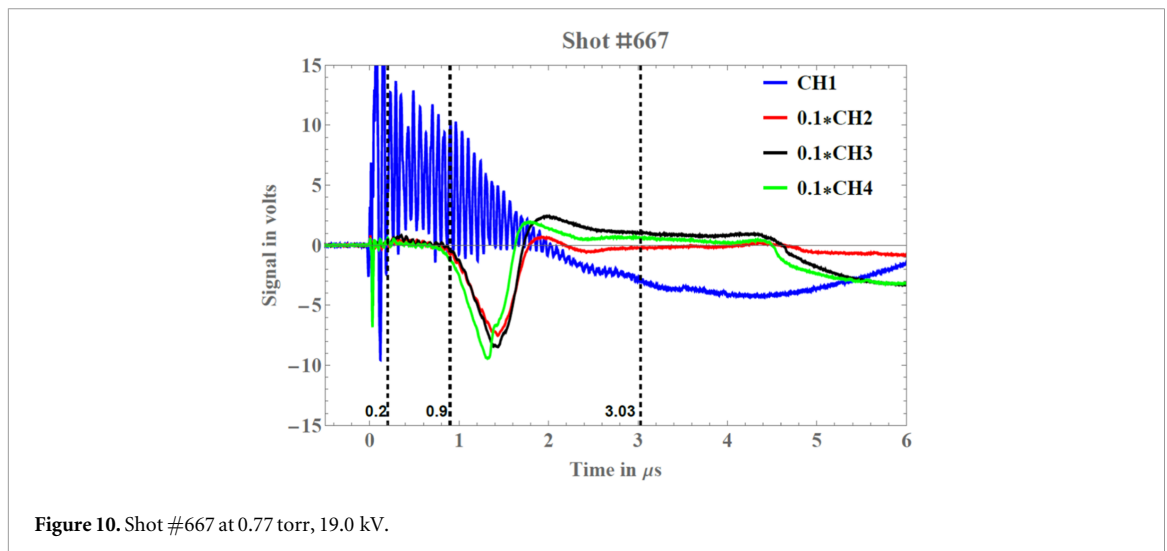
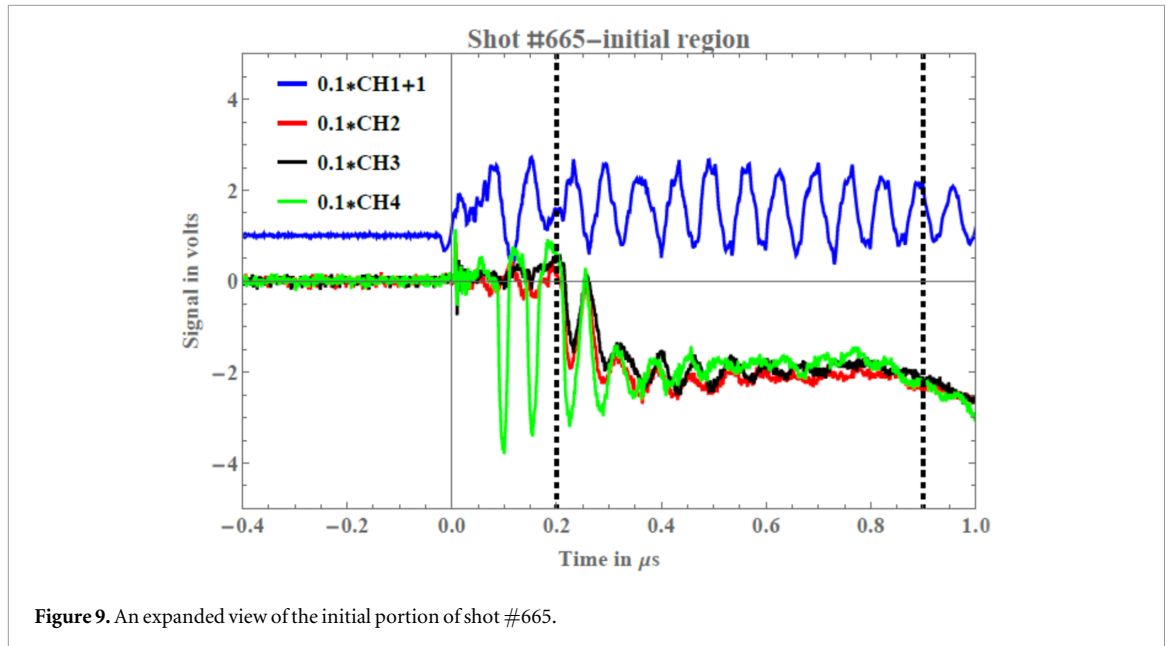


Figure 10 shows Shot #667 at 0.77 torr—lower pressure than #668.

The formation phase and the rundown phase look very similar to Shot #668, but the d-dot signals from CH2 and CH3 do not coincide as well. There is a barely observable change in the curvature of the current derivative signal slightly before the vertical dashed line at $3.03\ \mu\text{s}$ —the location of the singularity in Shot #668—suggesting that the singularity would have formed there had the symmetry been better in the rundown phase.

5. Summary and conclusion

This paper introduces a new diagnostic technique that is sensitive to the processes occurring in the formation phase of the Dense Plasma Focus—an aspect much neglected in plasma focus research but is of crucial importance for developing this versatile device into a technology platform. It measures the electric flux emitted from a plasma that is incident on a floating conductor which is part of the detector. The charge induced by the incident electric flux is carried to the electrical ground through a resistance, generating a diagnostic signal. A simple theory provides a relation between the electric flux and the measured signal. The value of the diagnostic, however, lies in its adaptation to the plasma sheath formation region, where three identical detectors look at the plasma symmetrically. It is shown that the signals have a similar shape and magnitude in the formation and rundown phases but differ considerably in the radial phase.

The paper demonstrates that this diagnostic provides information about the plasma formation phase—particularly its symmetry, duration and dynamics. It has enabled detection of a hidden hardware error, which went un-noticed for many months of operation. Two shots, both forming a pinch, are seen to exhibit different dynamics of the formation phase. Further studies would go a long way in understanding, modelling and control of the formation phase, improving prospects of developing industrial technologies based on the Dense Plasma Focus.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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